Unit 07 Practice Problems: Confidence Intervals

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Required Practice Problems: Confidence Intervals

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and/or preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas (`Unit-PP-07-Solution.pdf > Practice Problem-Solution > STAT 312 > Files`).

7.1

If someone wanted to compute a 76.2% confidence interval for μ when σ is known, what value should they use for $Z_{\alpha/2}$ while using the formula $\overline{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$?

7.2

If someone wanted to compute a 87.4% confidence interval for μ when σ is known, what value should they use for $Z_{\alpha/2}$ while using the formula $\overline{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$?

7.3

A population of moose lives in a national park. A random sample of $n = 35$ moose has a sample mean weight of $\overline{X} = 944$ pounds. Based on data from previous studies, the biologists assume σ is known to be 92 pounds.

(a) Compute a 90% confidence interval for μ = the population mean weight among all the moose living in the national park. What assumptions, if any, are required?

(b) What if the error margin is considered too large and an error margin of 20 pounds is desired at 90% confidence? What sample size is necessary?

7.4

An automotive engineer is interested in the population of all vehicles that have an engine of size of 2.0 L or smaller, and is particularly interested in μ = the population mean mpg (miles per gallon) among these vehicles. The sample mean among a random sample of $n = 12$ vehicles is 36.1 mpg and σ is known to be 4.4.

(a) Compute the 95% CI for μ. What assumptions, if any, are required?

(b) What if the error margin is considered too large and an error margin of 1.6 mpg is desired at 95% confidence? What sample size is necessary?

7.5

Draw pictures:

(a) On a T curve with 19 degrees of freedom, find the area to the right of 2.093.

(b) On a T curve with 8 degrees of freedom, find the area between -1.397 and 1.397.

(c) How many degrees of freedom does a T curve have if the area to the left of 2.763 equals 0.995?

(d) How many degrees of freedom does a T curve have if the area between -2.160 and 2.160 equals 0.95?

7.6

If someone wanted to compute a 98% confidence interval for μ when σ is unknown and $n = 23$, what value should they use for $t_{\alpha/2}$ while using the formula $\overline{X} \pm t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}$?

7.7

If someone wanted to compute a 90% confidence interval for μ when σ is unknown and $n = 13$, what value should they use for $t_{\alpha/2}$ while using the formula $\overline{X} \pm t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}$?

7.8

Nutrition scientists randomly select $n = 10$ students at a university and record how many calories they consume per day on average. The data set has a sample mean of $\overline{X} = 2681$ and a sample standard deviation of $s = 241$. σ is unknown. Compute a 95% confidence interval for $\mu =$ the population mean daily calorie consumption among all students at the university. What assumptions, if any, are required?

7.9

A medical researcher is studying how much weight women gain during pregnancy. μ = population mean weight gain during pregnancy. She randomly selects $n = 30$ women who recently gave birth and recorded how much weight they gained during pregnancy. The sample mean is $\overline{X} = 27.6$ pounds and the sample standard deviation is $s = 7.93$ pounds. σ is unknown. Compute a 90% confidence interval for μ. What assumptions, if any, are required?

7.10

In a large shipment of oranges, some oranges have seeds and some are seedless. You are interested in p = the proportion of oranges that are seedless. You randomly select 100 oranges and cut them open. 81 are seedless.

(a) Compute a 99% confidence interval for the proportion of oranges in the entire shipment that are seedless.

(b) What if the error margin is considered too large and an error margin of .03 is desired at 99% confidence? What sample size is necessary?

7.11

Forty high school students who recently earned their driver's license were randomly selected and 31 said they always wear a seat belt while driving.

(a) Use this data to compute a 95% confidence interval for p = the proportion of all recently licensed high school drivers who always wear a seat belt.

(b) What if the confidence level had been 90% instead of 95%? Would the confidence interval have been wider or thinner?

(c) What if 310 out of 400 high school students said they always wear a seat belt instead of 31 out of 40? Would the 95% confidence interval have been wider, thinner, or the same width as the one you computed in part a?

(d) What if the error margin is considered too large and an error margin of .09 is desired at 95% confidence? What sample size is necessary?

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.
Chapter 2 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 8.1.1 (a-c), 8.1.3, 8.1.5, 8.1.7 (a), 8.1.8, 8.1.13 (ac), 8.2.2, 8.2.3, 8.2.7, 8.2.9 (ab), 8.4.1 (a), 8.4.3, 8.4.5

[To find these exercises in the digital copy of the textbook (`Textbooks > STAT 312 > Files > Canvas`), go to the xvi Contents page and click the "EXERCISES" tab located at the lower-right corner of the page. This will open the complete list of chapter exercises. The **solutions** are posted on Canvas (`textbook-solution.pdf > Practice Problem-Solution > STAT 312 > Files`).]